

- Always prove correctness of your algorithms.
- You do not need to analyze the time complexity.

## Question 1 (7 points)

Let  $X = \{x_1, x_2, \dots, x_n\}$  be a set of  $n$  elements. Let  $S_1, S_2, \dots, S_m$  be subsets of  $X$ , i.e., each  $S_j \subseteq X$  with weights  $w_1, w_2, \dots, w_m$  respectively. Assume that every item in  $X$  appears in some set, i.e.,  $\bigcup_j S_j = X$ . A set cover of  $X$  with  $S$  is a set  $I \subseteq \{1, \dots, m\}$  such that  $\bigcup_{j \in I} S_j = X$ . A **minimum set cover** is a set cover  $I$  of minimum weight, i.e.,  $\sum_{j \in I} w_j$  is minimized.

- Write an Integer Linear Program for the minimum weighted set cover problem.
- If for every  $i \in [n]$ ,  $x_i$  appears in at most  $k$  many  $S_j$ 's then give a polynomial-time algorithm that finds a  $k$ -approximation for this bounded weighted set cover problem.

## Question 2 (7 points)

- Let  $f$  and  $f'$  be two feasible  $(s, t)$ -flows in a flow network  $G$ , such that  $|f'| > |f|$ . Prove that there is a feasible  $(s, t)$ -flow with value  $|f'| - |f|$  in the residual network  $G_f$ .
- Prove or disprove the following statement: Given a flow network  $G$ , if we replace a directed edge  $e$  with two directed edges  $e_1$  and  $e_2$  in opposite directions, and with the same capacity, then the value of the maximum flow remains unchanged.

## Question 3 (6 points)

In the *bin-packing problem*, we are given  $n$  pieces (or items) with specified sizes  $a_1, a_2, \dots, a_n$ , such that

$$1 > a_1 \geq a_2 \geq \dots \geq a_n > 0;$$

we wish to pack the pieces into bins, where each bin can hold any subset of pieces of total size at most 1, so our goal is to find a packing which minimize the number of bins used.

Find a polynomial time algorithm which gives us a packing which uses  $OUT$  bins, such that  $OUT \leq 2OPT + 1$ , where  $OPT$  is the minimum number of bins needed to pack the  $n$  items.

### Question 4 (7 points)

In the maximum 3-cut problem, we are given an undirected graph  $G = (V, E)$ , and nonnegative weights  $w_{ij} \geq 0$  for all  $(i, j) \in E$ . The goal is to partition the vertex set  $V$  into 3 parts,  $V_1$ ,  $V_2$ , and  $V_3$  so as to maximize the weight of all edges whose endpoints are in different parts (i.e.,  $\max_{(i,j) \in E: i \in V_a, j \in V_b, a \neq b} w_{ij}$ ). Give a deterministic polynomial time algorithm that finds a  $\frac{2}{3}$ -approximate solution for the maximum 3-CUT problem.

### Question 5 (6 points)

Assume you are given a connected, undirected graph  $G = (V, E)$  where every node has degree  $2k$ , for some integer  $k \geq 1$ . An orientation of  $G$  assigns a direction to every edge in  $E$ , i.e., for each edge  $(u, v)$ , it outputs either  $(u \rightarrow v)$  or  $(v \rightarrow u)$ . The result is a directed graph  $\text{orient}(G)$ .

Prove that there exists an orientation of the graph so that the directed graph  $\text{orient}(G)$  is strongly connected. (Recall that in a strongly connected graph, there is a directed path between every pair of nodes in both directions.)

### Question 6 (7 points)

Give a linear-programming formulation of the **bipartite maximum matching** problem. The input is a bipartite graph  $G = (U \cup V; E)$ , where  $E \subseteq U \times V$ ; the output is the largest matching in  $G$ . Your linear program should have one variable for each edge. Now dualize the linear program. What do the dual variables represent? What does the objective function represent? What problem is this!?

END OF PAPER